

Test 2 Set X

Physics

Date: August 27, 2026

Full Marks: 100

Class – XI

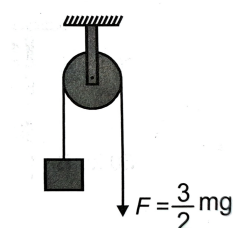
Time: 50.0 Minutes

General Instructions

1. Negative Marking is -1 for a single answer wrong
2. Write your Name and Roll Number clearly on the answer script.
3. Use blue or black ink only.
4. Mobile phones, calculators, and other electronic devices are not permitted.
5. There is only Section A but in your OMR fill the questions 21 to 25 under Section B.

SECTION A: Single Correct Answer Type

1. In the arrangement shown, the mass m will ascend with an acceleration (Pulley and rope are massless)
2. A rifle bullet loses $\frac{1}{20}$ th of its velocity in passing through a plank. Assuming that the plank exerts a constant retarding force, the least number of such planks required just to stop the bullet is



- (A) Zero (B) g
(C) $\frac{g}{2}$ (D) $2g$

- (A) 20 (B) Infinite
(C) 11 (D) 21
3. A ball of mass m moving with velocity u collides head-on with the second ball of mass m at rest. If the coefficient of restitution is e and velocity of first ball after collision is v_1 and velocity of second ball after collision is v_2 then

- (A) $v_1 = \frac{(1+e)u}{2}$, $v_2 = \frac{(1-e)u}{2}$
 (B) $v_1 = (1+e)u$, $v_2 = (1-e)u$
 (C) $v_1 = \frac{(1-e)u}{2}$, $v_2 = \frac{(1+e)u}{2}$
 (D) $v_1 = \frac{u}{2}$, $v_2 = -\frac{u}{2}$

4. The position x of a particle moving along x-axis at time t is given by the equation $t = \sqrt{x} + 2$, where x is in metres and t in seconds. Find the work done by the force in first four seconds.

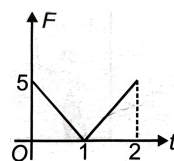
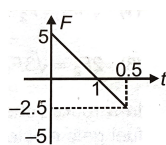
- (A) 8 J (B) Zero
 (C) 2 J (D) 4 J

5. A particle of mass $2m$ moving with velocity v strikes a stationary particle of mass $3m$ and sticks to it. The speed of the system will be

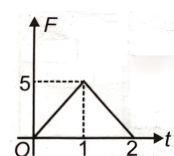
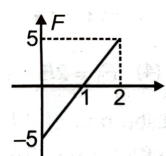
- (A) $0.2v$ (B) $0.4v$
 (C) $0.8v$ (D) $0.6v$

6. In which of the following graphs, the total change in momentum is zero?

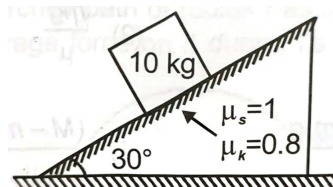
- (A) (B)



- (C) (D)



7. A block of 10 kg mass is placed on a rough inclined plane as shown in figure. The acceleration of the block will be



- (A) $\frac{\sqrt{3}g}{2}$ (B) $\frac{g}{2}$
 (C) g (D) Zero

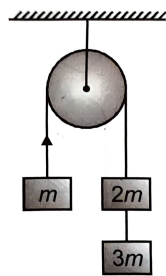
8. In accordance with Newton's third law of motion

- (A) All of these
 (B) This law is applicable whether the bodies are at rest or they are in motion
 (C) For appearance of action and reaction, physical contact is not necessary
 (D) Action and reaction never balance each other

9. If a force of constant magnitude acts in direction perpendicular to the motion of a particle, then its

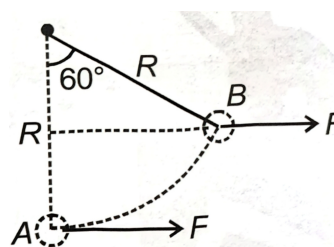
- (A) Momentum is uniform
 (B) Velocity is uniform
 (C) Speed is uniform
 (D) All of these

10. A string tied on a roof can bear a maximum tension of 50 kg wt. The minimum acceleration that can be acquired by a man of 98 kg to descend will be [Take $g = 9.8 \text{ m/s}^2$]
- (A) 4.8 m/s^2
 (B) 5 m/s^2
 (C) 9.8 m/s^2
 (D) 4.9 m/s^2
11. A 10 g bullet moving at 200 m/s stops after penetrating 5 cm of wooden plank. The average force exerted on the bullet will be
- (A) -2000 N
 (B) 4000 N
 (C) -4000 N
 (D) 2000 N
12. In the figure given below, with what acceleration does the block of mass m move? (Pulley and strings are massless and frictionless)
13. The power of water pump is 4 kW. If $g = 10 \text{ m/s}^2$, the amount of water it can raise in 1 minute to a height of 20 m is
- (A) 2000 litre (B) 1200 litre
 (C) 1000 litre (D) 100 litre
14. A uniform chain of length L and mass M is lying on a smooth table and one third of its length is hanging vertically down over the edge of the table. If g is acceleration due to gravity, the minimum work required to pull the hanging part of the chain on the table is
- (A) $\frac{MgL}{18}$ (B) $\frac{MgL}{3}$
 (C) MgL (D) $\frac{MgL}{9}$
15. An unloaded bus can be stopped by applying brakes on straight road after covering a distance x . Suppose, the passenger add 50
- (A) $2x$ (B) 0
 (C) $1.5x$ (D) $2.5x$



- (A) $\frac{g}{3}$ (B) $\frac{2g}{5}$
 (C) $\frac{2g}{3}$ (D) $\frac{g}{2}$
16. A block of mass m is at rest on a rough inclined plane of angle of inclination θ . If coefficient of friction between the block and the inclined plane is μ , then the minimum value of force along the plane required to move the block on the plane is
- (A) $mg[\sin \theta + \mu \cos \theta]$
 (B) $mg[\sin \theta - \mu \cos \theta]$
 (C) $mg[\mu \cos \theta + \sin \theta]$
 (D) $mg[\mu \cos \theta - \sin \theta]$

17. A machine gun fires a bullet of mass 65 g with a velocity of 1300 m/s. The man holding it can exert a maximum force of 169 N on the gun. The number of bullets he can fire per second will be
- (A) 3 (B) 2
(C) 4 (D) 1
18. A spring of force constant K is first stretched by distance a from its natural length and then further by distance b . The work done in stretching the part b is
- (A) $\frac{1}{2}Ka(a+b)$
(B) $\frac{1}{2}Kb(a-b)$
(C) $\frac{1}{2}Ka(a-b)$
(D) $\frac{1}{2}Kb(2a+b)$
19. A U-238 nucleus, originally at rest, decays by emitting an α -particle, say with a velocity of v m/s. The recoil velocity (in ms^{-1}) of the residual nucleus is
- (A) $-\frac{4v}{234}$ (B) $-\frac{4v}{238}$
(C) $\frac{4v}{238}$ (D) $\frac{v}{4}$
20. A stationary particle explodes into two particles of masses x and y , which move in opposite directions with velocity v_1 and v_2 . The ratio of their kinetic energies ($E_1 : E_2$) is
- (A) $\frac{y}{x}$ (B) $\frac{x}{y}$
(C) $\frac{xv_2}{yv_1}$ (D) 1
21. KE acquired by a mass m in travelling a certain distance d , starting from rest, under the action of a constant force F is
- (A) Directly proportional to $\sqrt{m}$
(B) Directly proportional to m
(C) Directly proportional to $\frac{1}{m}$
(D) None of these
22. Newton's first law is applicable
- (A) Only in inertial reference frames
(B) In all reference frames
(C) Only in non-inertial reference frames
(D) None of these
23. A block of mass m is pulled along a circular arc by means of a constant horizontal force F as shown. Work done by this force in pulling the block from A to B is



- (A) $\frac{FR}{2}$ (B) mgR
(C) FR (D) $\frac{\sqrt{3}}{2}FR$

24. A rocket of mass 5700 kg ejects mass at a constant rate of 15 kg/s with constant speed of 12 km/s. The acceleration of the rocket 1 minute after the blast is ($g = 10 \text{ m/s}^2$)
- (A) 27.5 m/s^2 (B) 34.9 m/s^2
(C) 3.50 m/s^2 (D) 13.5 m/s^2
25. A spring with spring constant k when compressed by 1 cm, the potential energy stored is U . If it is further compressed by 3 cm, then change in its potential energy
- (A) $15U$ (B) $3U$
(C) $8U$ (D) $9U$